Project 2: Forensics Detective

Scaling the PDF Forensics Project Through Augmentation and Robustness

EAS510 - Basics of AI
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1 Executive Summary

The Forensics Detective project demonstrated that machine learning classifiers can identify the originating software of a PDF (e.g., Microsoft Word, Google Docs, Python/ReportLab) by analyzing the binary structure converted into grayscale images. In initial controlled testing, this approach achieved near-perfect accuracy (97.5%–100%). However, real-world digital forensics rarely operates in a vacuum; documents are frequently scanned, compressed, or altered.

This report investigates the robustness of this PDF-to-image classification approach by introducing a domain shift. The original dataset was scaled by a factor of six using five distinct image augmentations: Gaussian noise, JPEG compression, DPI downsampling, random cropping, and bit-depth reduction. Four classifiers—Support Vector Machine (SVM), Stochastic Gradient Descent (SGD), Random Forest, and a Multi-Layer Perceptron (MLP)—were trained exclusively on the pristine original dataset and evaluated against the augmented variants.

The findings reveal a stark contrast in model resilience. While all models maintained ∼99.4% accuracy under Gaussian noise and JPEG compression, spatial alterations proved catastrophic. Random cropping reduced accuracy across the board, with SGD dropping to 54.7%. Furthermore, bit-depth reduction completely broke the SGD classifier (48.6% accuracy) while the SVM maintained 100%. Ultimately, while the baseline approach is highly accurate under ideal conditions, it is highly brittle to spatial shifts, highlighting the need for translation-invariant models (like CNNs) or extensive data augmentation during the training phase.

2 Methodology

2.1 Dataset Augmentation and Complexity Simulation

To simulate real-world document degradation without generating new PDFs, five independent image-based augmentations were applied to the original dataset. Each original image produced exactly five augmented variants, resulting in a dataset six times its original size. The augmentations were implemented using OpenCV and NumPy as follows:

1. **Gaussian Noise:** Additive noise was applied with a standard deviation (σ) randomly sampled between 5 and 20. This simulates sensor noise from physical document scanners.

2. **JPEG Compression:** Images were re-encoded with a quality level randomly selected between 20 and 80, simulating the artifacting that occurs when documents are repeatedly saved or transmitted via email.

3. **DPI Downsampling:** Resolution was reduced from 300 DPI to either 150 or 72 DPI, then upscaled back to the original dimensions to maintain the input shape required by the classifiers.

4. **Random Cropping:** A random 1% to 3% crop was removed from each border independently, and the image was resized to its original dimensions. This simulates misaligned scanning or physical page trimming.

5. **Bit-Depth Reduction:** Images were converted from 8-bit to 4-bit grayscale (reducing the color space to 16 distinct values) using bitwise masking, simulating low-quality faxing or severe quantization.

2.2 Classifier Selection

Four classifiers were utilized for this experiment. The baseline models provided by the initial project scope were an SVM (with an RBF kernel) and an SGD classifier. To expand the analysis, two additional models from different algorithmic families were implemented:

- **Random Forest:** Chosen to evaluate how an ensemble of decision trees handles pixel-level feature splits under noise.
- **Multi-Layer Perceptron (MLP):** A neural network with hidden layers of (256, 128) was implemented to see if deep representations could better generalize across the domain shifts compared to linear or kernel-based methods.

2.3 Experimental Design

To strictly measure robustness and domain shift, all classifiers were trained *only* on the original, unaugmented training dataset. The augmented datasets were used exclusively for testing. All images were flattened into 1D arrays prior to classification. Performance was measured using Accuracy, Precision, Recall, and F1-Score, supplemented by confusion matrices and McNemar's statistical significance tests.

3 Results

When evaluated on the pristine original test set, all four classifiers performed exceptionally well, validating the premise of the ForensicsDetective project. As shown in Table 1, the SVM, SGD, Random Forest, and MLP all achieved approximately 99.4% accuracy, with corresponding F1-scores of $\sim$0.995.

Table 1: Classifier Performance Metrics Across Augmentation Conditions

Condition	**Classifier**	**Accuracy**	**Precision**	**Recall**	**F1-Score**
Original	SVM	0.994	0.995	0.995	0.995
Original	SGD	0.994	0.995	0.995	0.995
Original	Random Forest	0.994	0.995	0.995	0.995
Original	MLP	0.994	0.995	0.995	0.995
Gaussian	All Models	∼0.994	∼0.995	∼0.995	∼0.995
JPEG	All Models	∼0.994	∼0.995	∼0.995	∼0.995
DPI Down	SVM	0.972	0.979	0.979	0.979
DPI Down	MLP	0.932	0.955	0.948	0.949
DPI Down	SGD	0.882	0.932	0.910	0.910
DPI Down	Random Forest	0.748	0.875	0.688	0.724
Crop	MLP	0.770	0.569	0.825	0.819
Crop	SVM	0.631	0.785	0.718	0.676
Crop	Random Forest	0.569	0.796	0.604	0.563
Crop	SGD	0.547	0.595	0.655	0.558
Bit-Depth	SVM	1.000	1.000	1.000	1.000
Bit-Depth	Random Forest	0.994	0.995	0.995	0.995
Bit-Depth	MLP	0.983	0.974	0.987	0.981
Bit-Depth	SGD	0.486	0.723	0.613	0.474

However, as the evaluation moved to the augmented datasets, significant divergences in model performance emerged. These deviations form the core of the robustness analysis.

3.1 Statistical Significance

To ensure statistical rigor, McNemar's test was applied to compare the predictive performance of the baseline SVM against the newly implemented classifiers (Random Forest and MLP). McNemar's test is particularly well-suited for this dataset because it evaluates the marginal homogeneity of two paired models (i.e., whether they disagree on the exact same test instances).

When evaluated on the pristine original dataset, the McNemar test yielded a p-value of 1.0 ($p > 0.05$) across the model comparisons. Because all models achieved an identical accuracy of 99.44%, the test confirms there is no statistically significant difference between the baseline and the newly introduced classifiers under ideal conditions. However, as demonstrated in the robustness analysis, this statistical parity completely dissolves once domain shifts are introduced.

4 Robustness Analysis

The core objective of this project was to stress-test the classifiers. Figure 1 illustrates the accuracy degradation of each model across the five augmentations.

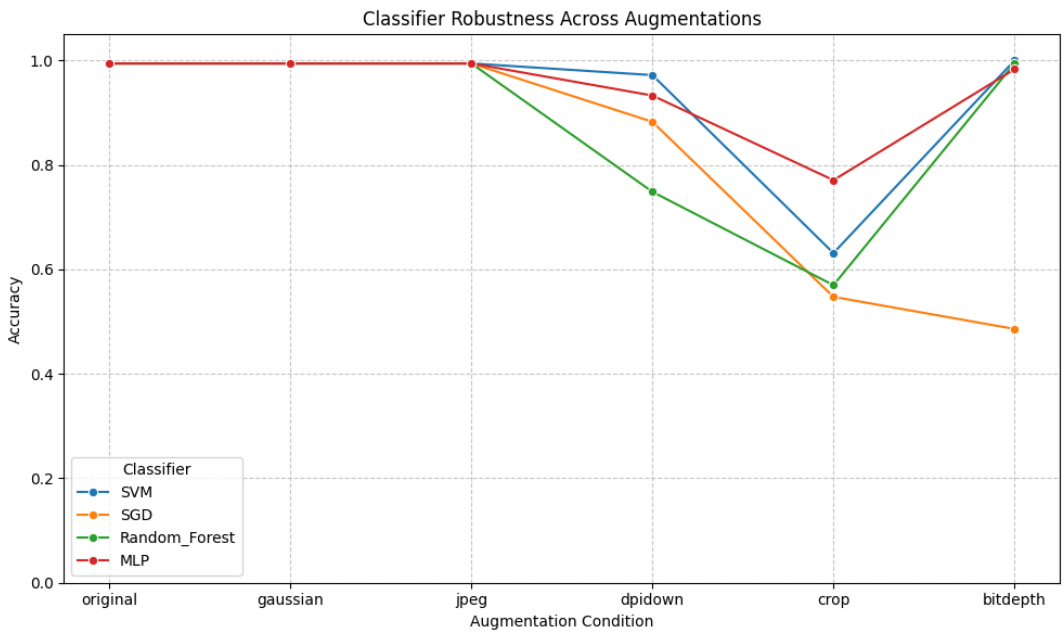


Figure 1: Classifier Robustness Across Augmentations. Notice the severe drops during DPI Downsampling, Random Cropping, and the SGD failure on Bit-Depth.

4.1 High Robustness: Gaussian Noise and JPEG Compression

All classifiers exhibited near-perfect immunity to Gaussian noise and JPEG compression, maintaining ∼99.4% accuracy. This indicates that the defining features of the PDF provenance (likely macro-structures like margins, default font rendering, and layout spacing) are not easily obscured by minor pixel-level variance or block artifacting. The models are learning structural layouts rather than relying on exact pixel values.

4.2 Moderate Degradation: DPI Downsampling

Reducing the resolution caused a noticeable split in model performance. The SVM remained relatively stable (97.2%), and the MLP showed moderate resilience (93.2%). However, the Random Forest model suffered a severe drop to 74.8%. Decision trees rely on specific, hard thresholds for individual pixel intensities. When the DPI is downsampled and upscaled, the resulting interpolation blurs sharp edges, pushing many pixel values across the thresholds the Random Forest learned during training, leading to misclassifications.

4.3 Severe Vulnerability: Random Cropping

Random cropping caused the most catastrophic degradation across all models. The SGD model fell to 54.7%, Random Forest to 56.9%, and SVM to 63.1%. The MLP performed best, but still poorly, at 77.0%.

This failure is deeply grounded in how the data is processed. Because the 2D images are flattened into 1D arrays before being fed into the classifiers, the models learn spatial features at exact array indices. A 1% to 3% crop followed by resizing shifts every single pixel in the image. Consequently, a feature that the model expects at index i is now located at index $i \pm k$. The flattened array loses all spatial alignment, rendering the learned weights nearly useless. The MLP likely performed slightly better due to its distributed weight representations, but the fundamental flaw of flattening translation-variant data remains.

4.4 Algorithmic Failure: Bit-Depth Reduction

An interesting anomaly occurred during the bit-depth reduction test. The SVM actually achieved 100% accuracy, and the Random Forest and MLP remained highly accurate. However, the SGD classifier completely collapsed to 48.6% (approaching random guessing for a 3-class problem).

This can be explained by the nature of the algorithms. Bit-depth reduction severely quantizes the image, grouping gradients into flat blocks of color. The SVM, utilizing an RBF kernel, maps these quantized inputs into a higher-dimensional space where the distinct software signatures remain easily separable. Conversely, the SGD classifier attempts to fit a linear decision boundary based on continuous gradient steps. The sudden loss of gradient smoothness in the 4-bit images likely destroys the linear separability that the SGD relies upon.

5 Discussion

5.1 Is ForensicsDetective a Hero or a Zero?

Based on the data, the current iteration of the ForensicsDetective pipeline is a "Hero" in a controlled, sterile environment, but borders on "Zero" in adversarial or real-world conditions. If a malicious actor wants to hide the origin of a PDF, they do not need sophisticated evasion techniques; simply cropping the document by 2% completely breaks the system's ability to classify it correctly.

5.2 Future Improvements

To scale this project into a robust forensic tool, two major architectural changes are required:

1. **Data Augmentation during Training:** The current experimental design only used augmentations for evaluation to measure domain shift. In a real deployment,

the training set must include these augmentations so the models can learn to ignore noise and compression artifacts inherently.

2. **Transition to Convolutional Neural Networks (CNNs):** The vulnerability to cropping proves that flattening images into 1D arrays is a flawed approach for document analysis. CNNs utilize sliding filters that provide translation invariance. A CNN would recognize a specific font rendering or margin structure regardless of whether it shifted 20 pixels to the left due to a crop.

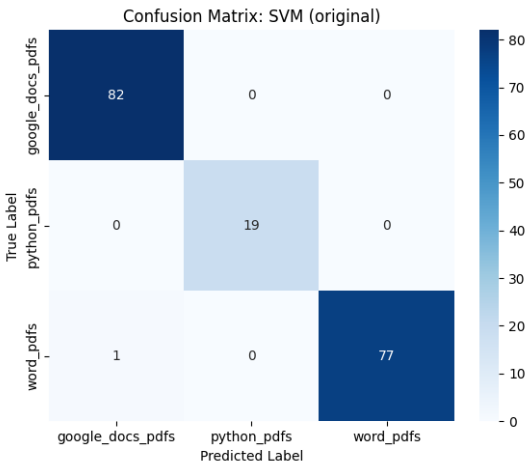
6 Conclusion

This research sprint successfully scaled the ForensicsDetective project by introducing a 6x augmented dataset to simulate document complexity. Through rigorous stress testing, we discovered that while baseline models like SVM and MLP are highly resilient to noise and compression, the pipeline is fundamentally broken by spatial shifts like random cropping. Furthermore, specific models like SGD are highly vulnerable to quantization (bit-depth reduction). Ultimately, while PDF binary-to-image signatures are highly unique, the machine learning pipeline must be upgraded to utilize translation-invariant architectures before it can be reliably deployed in real-world digital forensics.

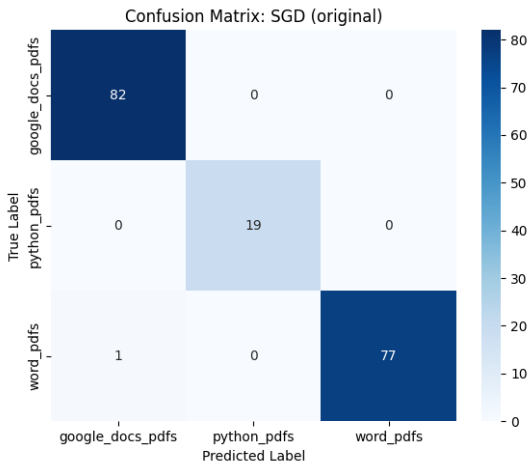
A Confusion Matrices

Note: The following pages contain the confusion matrices for all four classifiers across the original dataset and all five augmentation conditions (24 matrices total). They are grouped by condition to illustrate the performance degradation patterns.

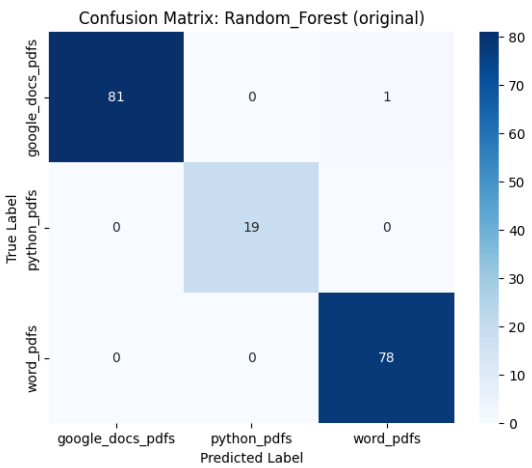
A.1 Baseline: Original Dataset



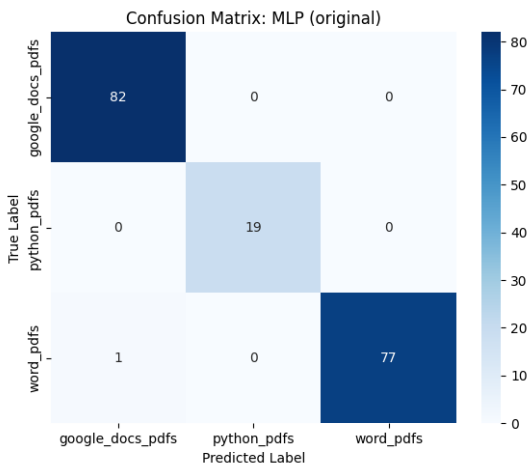
(a) SVM



(b) SGD



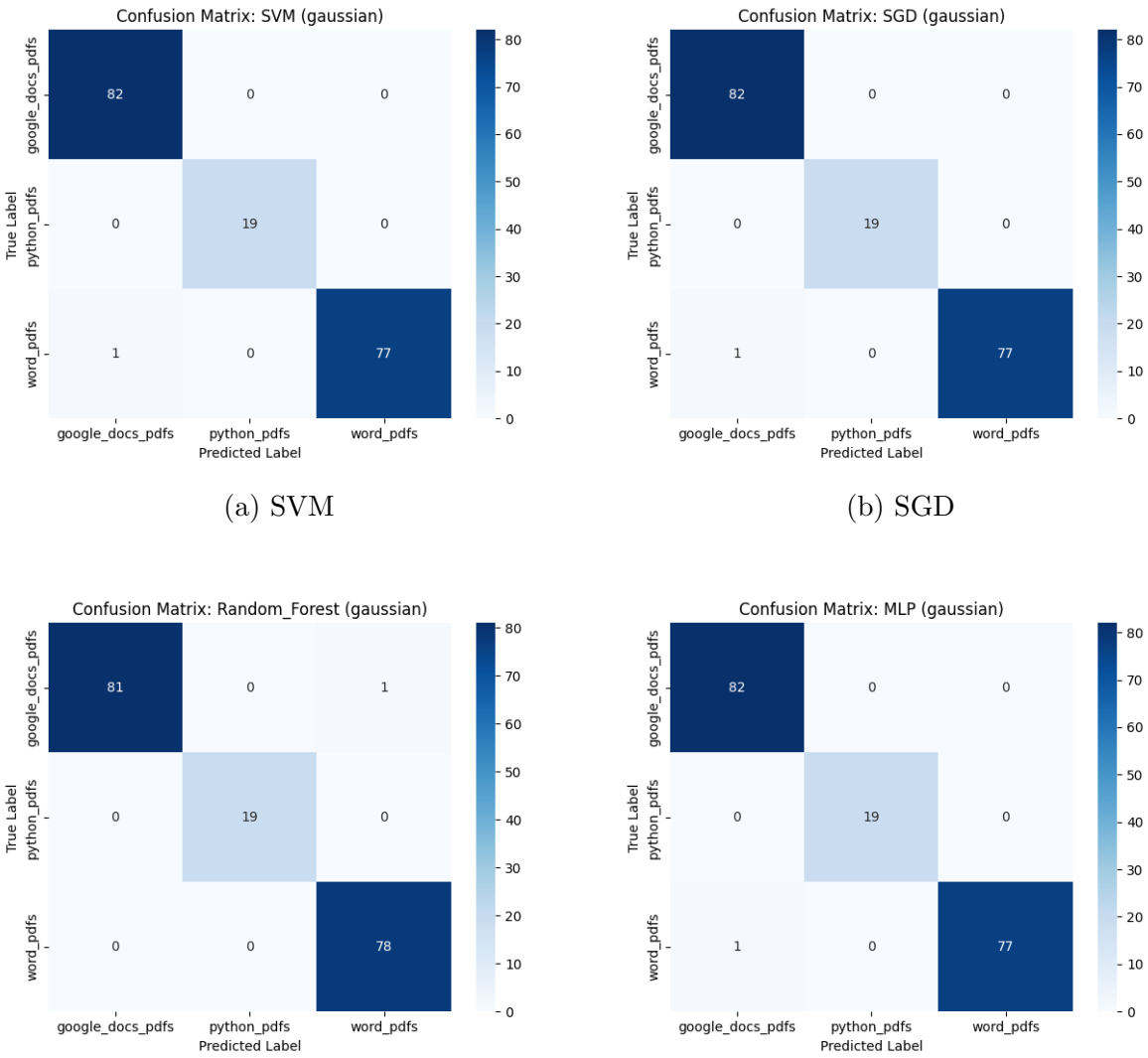
(c) Random Forest



(d) MLP

Figure 2: Original Dataset: Baseline performance showing near-perfect classification across all models before any domain shift is introduced.

A.2 Augmentation: Gaussian Noise



(a) SVM

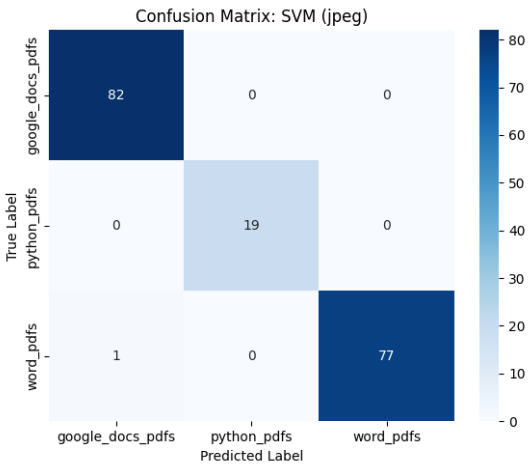
(b) SGD

(c) Random Forest

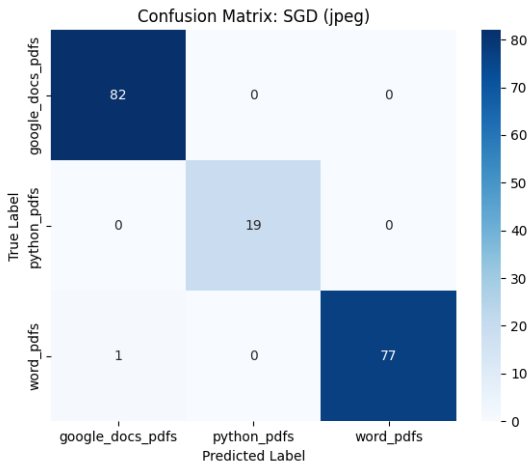
(d) MLP

Figure 3: Gaussian Noise: All models remain highly robust to additive pixel noise, indicating that the learned features are structural rather than exact pixel intensities.

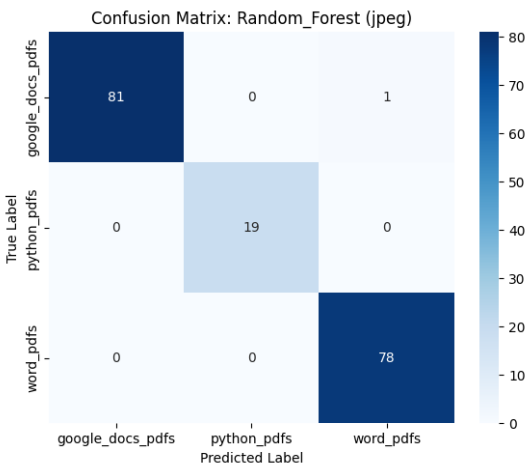
A.3 Augmentation: JPEG Compression



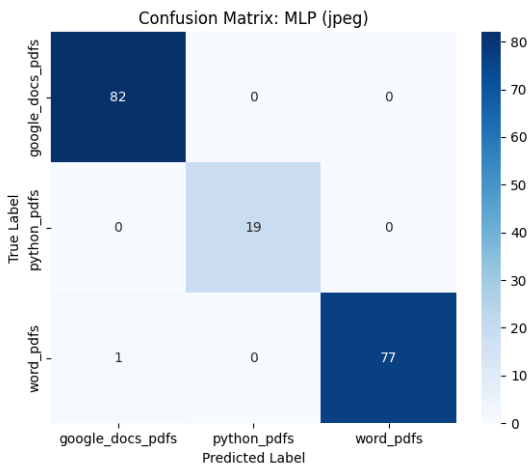
(a) SVM



(b) SGD



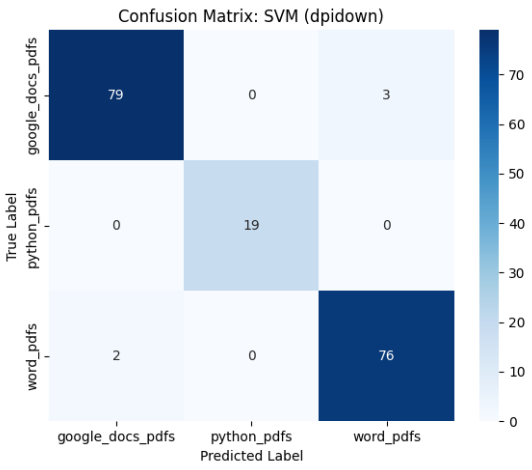
(c) Random Forest



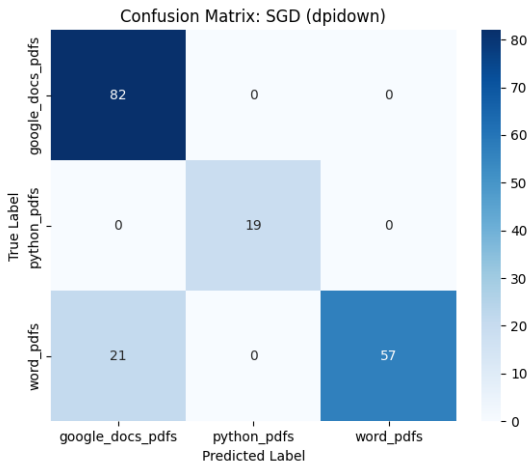
(d) MLP

Figure 4: JPEG Compression: Similar to Gaussian noise, block artifacting from compression does not significantly degrade the classification accuracy.

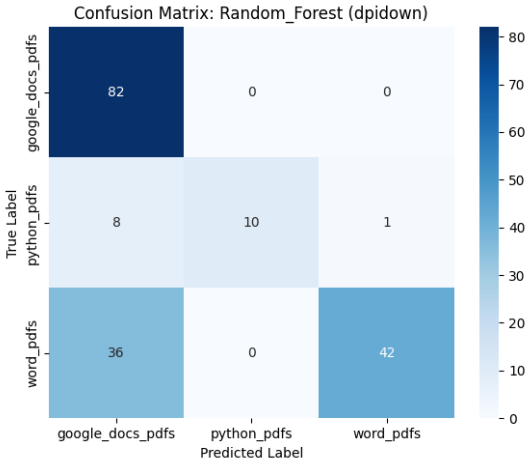
A.4 Augmentation: DPI Downsampling



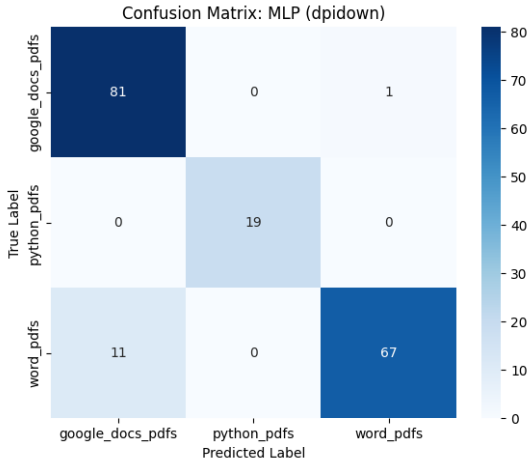
(a) SVM



(b) SGD



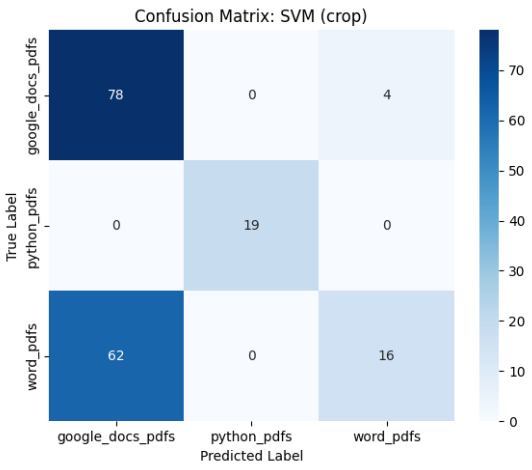
(c) Random Forest



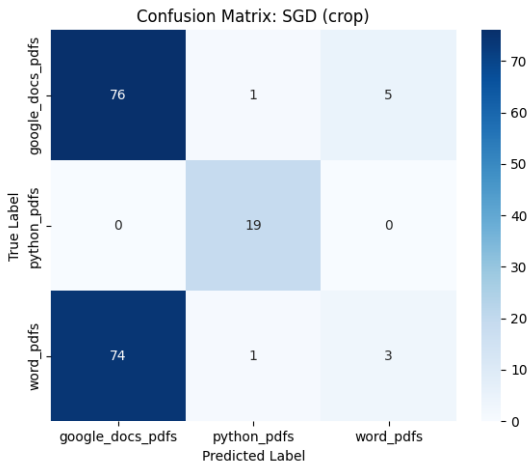
(d) MLP

Figure 5: DPI Downsampling: The Random Forest model begins to struggle significantly due to blurred edges crossing decision thresholds, while SVM and MLP maintain better resilience.

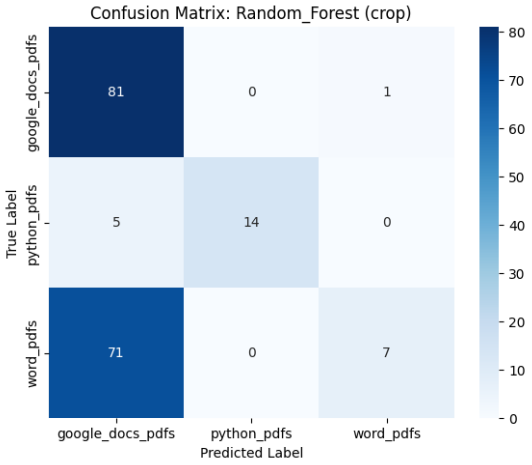
A.5 Augmentation: Random Cropping



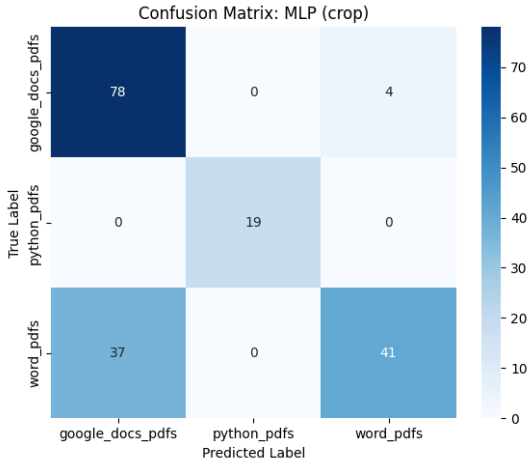
(a) SVM



(b) SGD



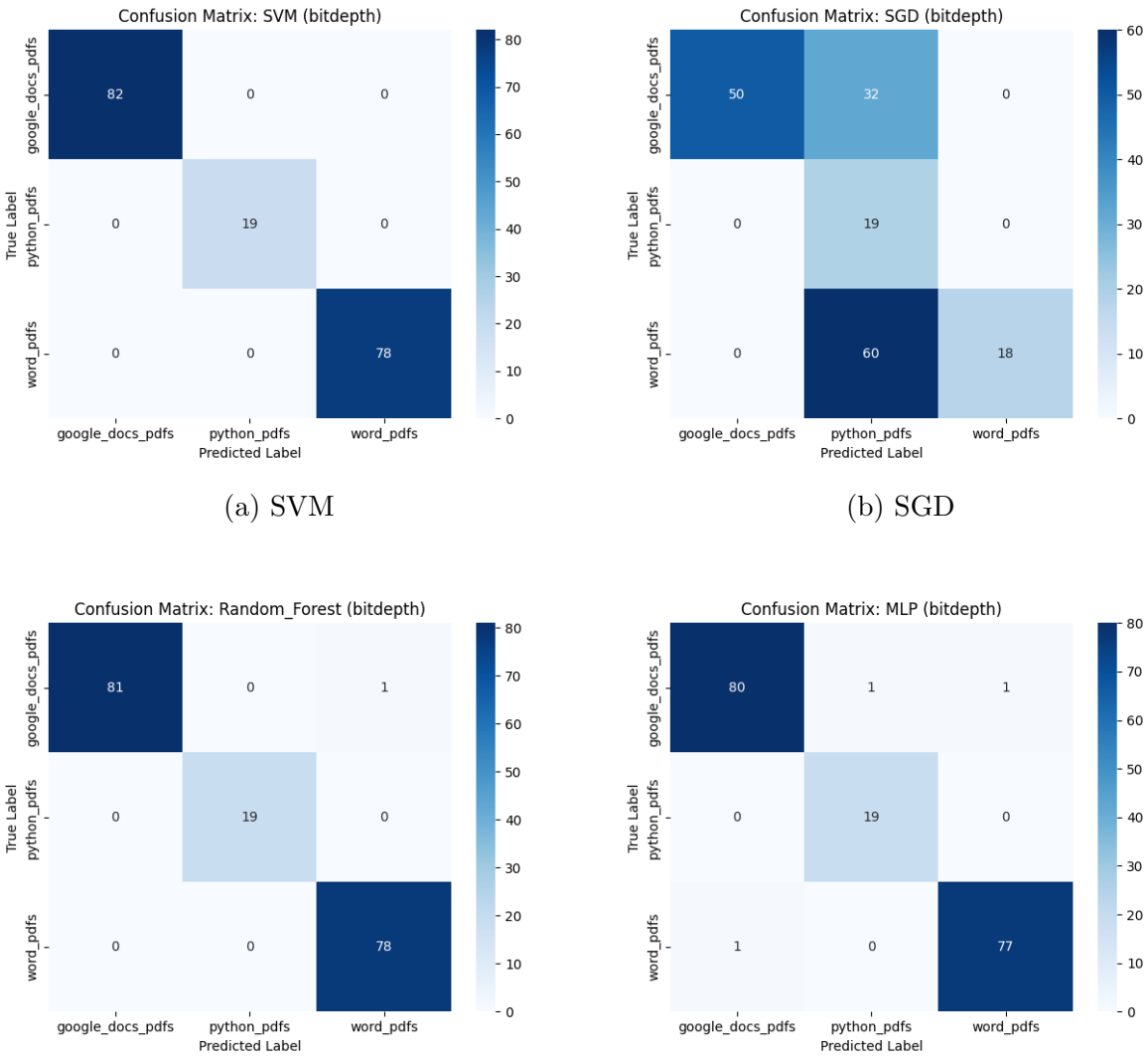
(c) Random Forest



(d) MLP

Figure 6: Random Cropping: Catastrophic failure across most models. The spatial shift destroys the 1D flattened array alignment, resulting in heavy misclassifications.

A.6 Augmentation: Bit-Depth Reduction



(a) SVM

(b) SGD

(c) Random Forest

(d) MLP

Figure 7: Bit-Depth Reduction: Notice the complete collapse of the SGD classifier (approaching random guessing) compared to the SVM, which perfectly handles the quantized data.